Elton Chang

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EDUCATION

Carnegie Mellon University May 2026
Master of Science in Data Analytics for Science, GPA: 3.97/4.0 Pittsburgh, PA

University of California, Santa Cruz Jun 2025
Bachelor of Science in Computer Science, GPA: 3.7/4.0 Santa Cruz, CA

EXPERIENCE

Co-Founder 2026 - Present
AMOS
- Co-developed a Rust prototype for an AI analyst that enforces data-access permissions and executes SQL and statistical calculations outside the language model.
- Implemented schema and metric checks, evidence-linked claims, human review, and replay to make analytical results traceable and inspectable.

Founder / Technical Lead Aug 2025 - Present
Ashe System Pittsburgh, PA / San Francisco, CA
- Improved XGBoost aircraft-on-ground risk ranking (Precision@20: 35% to 62%); tuned parameters and thresholds with Optuna, increasing median warning lead time from 6.5 to 10.2 days.

Infrastructure Engineer Intern Jul 2023 - Sep 2023
Wintec Industries Inc. Newark, CA
- Standardized testing environments across 15+ hosts with Terraform; built telemetry and alerting for backup and replication pipelines, reducing anomaly detection time by 40%.

Software Engineer Intern Jul 2022 - Sep 2022
Wintec Industries Inc. Newark, CA
- Automated file ingestion, metadata indexing, and lifecycle tracking in a cloud-hosted file management system using scheduled Python jobs.

PROJECTS

Next Best Action | CMU/PNC Capstone
- Compared HistGradientBoosting, LightGBM, XGBoost, and ensembles for next-month product adoption; the project achieved 0.811 ROC AUC and a 95% top-three hit rate on supported recommendations.

TraceFrame | Data Quality and Reproducibility
- Built Python tooling for pandas, Polars, and DuckDB analyses to track datasets, transformations, SQL outputs, metrics, and source-row lineage.
- Added schema, null, uniqueness, and range checks; generated charts and HTML audit reports that connect analytical claims to supporting data.

Quantifying Meritocracy | Agent-Based Policy Simulation
- Extended the Talent vs. Luck model with solidarity mechanisms to compare redistribution strategies using inequality, upward mobility, and wealth concentration metrics.
- Ran reproducible simulations and sensitivity analyses across policy timing and model parameters; documented results as model-based comparisons of policy scenarios.

Tactile Prediction and Contact Mechanics Oct 2026 - Present
- Built Python/PyTorch prediction models and retrieval/regression baselines; reproduced 1,878 sensor windows and 22 prediction arrays from recorded tactile data.
- Designed surface-level holdouts, controls, and paired bootstrap intervals; compared eight force-prediction methods and controls on 450 synthetic trials across 10 groups.

TECHNICAL SKILLS

Proficient Languages: Python, SQL, Rust, C++
Proficient Tools: PyTorch, scikit-learn, pandas, Polars, DuckDB, NumPy, SciPy, ROS 2